Total No. of Questions : 4]

SEAT No. :

PE25

[Total No. of Pages : 2

[6579]-325

T.E. (Computer Engineering/Computer Science/Artificial Intelligence and Data Science) (Insem)
DATABASE MANAGEMENT SYSTEMS
(2019 Pattern) (Semester - I) (310241)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) For the database system to be usable, it must retrieve data efficiently. The need of efficiency has led designers to use complex data structures to represent data in the database. Developers hide this complexity from the database system users through several levels of abstraction. Explain those levels of abstraction in detail. **[5]**

b) What is Specialization and generalization in Extended E-R diagram? What is the difference between them? Why do we not display the difference in schema diagram? **[6]**

c) Explain with example what is physical data independence. Also explain its importance. **[4]**

OR

Q2) a) Draw the neat diagram of Database System Structure and explain its components in detail. **[8]**

b) A post office has few postmen who go every day to distribute letter. Every morning post office receives a large number of registered letters. the post office intends to create a database to keep track of these letters. **[7]**

i) Every letter has a sender, an origin post office from where it was sent, a destination post office to which it is to be sent, a date of registration, date of arrival at destination post office, receiver and a status.

ii) Every sender has a name, an address.

iii) Every receiver has a name, an address.

P.T.O.

iv) Every postman has a designated area where he delivers letters.

v) The area consists of a set of streets under the jurisdiction of the post office.

vi) Every street consists of a set of buildings.

vii) Every building has number and may be name. It may be housing more than one family.

viii) The status of the letter can be not yet taken for delivery, delivered, address not available, address not known, addressee did not accept the letter, redirected to the address of address and sent to the sender.

Identify the relationship among the entities along with the mapping cardinalities, keys in the E.R. diagram. Construct appropriate tables for E-R diagram designed with above requirements

Q3) a) Consider the following schemas **[6]**

Emp(Emp_no, Emp_name, Dept_no)

Dept(Dept_no, Dept_name)

Address(Dept_name, Dept_location)

Write SQL queries for the following

i) Display the location of department where employee 'Ram' is working.

ii) Create a view to store total no of employees working in each department in ascending order

iii) Find the name of the department in which no employee is working.

b) Explain the concept of Referential and Entity Integrity constraint with example. **[5]**

c) Explain cursor in SQL with example. **[4]**

OR

Q4) a) What is trigger? How to create it? Discuss various types of triggers. **[5]**

b) Write a PL/SQL block of code which accepts the rollno from user. The attendance of rollno entered by user will be checked in student_attendance (RollNo, Attendance) table and display on the screen. **[5]**

c) What is sequence? How to create and use sequence in SQL? **[5]**

Total No. of Questions : 4]

SEAT No. :

PC-25

[Total No. of Pages : 2

[6360]-25

T.E. (Computer Engg) (Artificial Intelligence and Data Science)

DATABASE MANAGEMENT SYSTEM

(2019 Pattern) (310241) (Semester - I) (Insem)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Assume Suitable data, if necessary.***

Q1) a) Compare DBMS and File processing system in terms of data isolation, data redundancy, data inconsistency, data integrity. **[4]**

b) A reputed general hospital has decided to computerize their operations. In the hospital many doctors are working. Personal information of doctors is maintained to get them fixed salary per month. The patients are admitted to the hospital into the room. They are treated by various doctors. Sometimes patients perform certain pathological tests which carried out into the labs. Identify the relationship among the entities along with the mapping cardinalities, keys in the E.R. diagram. Construct appropriate tables for E-R diagram designed with above requirements. **[7]**

c) A weak entity set can always be made into a strong entity set; by adding to its attributes the primary key attributes of its identifying entity set. Outline what sort of redundancy will result if we do so while converting to tables. **[4]**

OR

Q2) a) What is Specialization and generalization in Extended E-R diagram? What is the difference between them? Why do we not display the difference in schema diagram? **[7]**

b) Draw architecture of DBMS system and explain function of following components: **[8]**

i) Storage manager

ii) Query Processor

P.T.O.

Q3) a) Consider following schema **[4]**

Student_fee_details (rollno, name, fee_deposited, date)

Write a trigger to preserve old values of student fee details before updating in the table.

b) What is view and how to create it? Can you update view? If yes, how? If not, why not? **[6]**

c) Consider following schema **[5]**

Hotels(hotel_no,hotel_name.city)

Rooms(Room_no,hotel_no,price,type)

Write a PL/SQL procedure to list the price & type of all rooms at the hotel 'TAJ'

OR

Q4) a) Consider the following relation schema **[6]**

MOVIES(Mov_Id, Mov_Title,Mov_Year, Dir_Id)

DIRECTOR(Dir_Id, Dir_Name)

RATING(MOV_Id, Rev_Stars)

Write the SQL queries for the following.

i) List the title of all the movies directed by 'RAJ KAPOOR'

ii) Find the name of movies and number of stars for each movie. Sort the results on movies title and from higher stars to least stars.

iii) Assign the rating of all movies directed by 'Steven Spielberg' to 9.

b) What is the importance of creating constraints on the table? Explain with example any 4 constraints that can be specified when a database table is created. **[5]**

c) What is synonym? How to create and use synonym in SQL? **[4]**

Total No. of Questions : 4]

SEAT No. :

P8555

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Oct-22/TE/Insem-525
T.E. (Computer Engg.) (AI & DSE)
DATABASE MANAGEMENT SYSTEM
(2019 Pattern) (Semester - I) (310241)

Time : 1 Hour] ***[Max. Marks : 30***

Instructions to the candidates :

1) ***Answer Q.1 or Q.2, Q.3 or Q.4.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) For the database system to be usable, it must retrieve data efficiently. The need of efficiency has led designers to use complex data structures to represent data in the database. Developers hide this complexity from the database system users through several levels of abstraction. Explain those levels of abstraction in detail with example. **[5]**

b) Draw an ER diagram for the banking system. Assume the banking requirements are as given below. **[10]**

- The bank is organized into branches. Each branch is located in a particular city.
- The bank offers two types of accounts: saving and current. Accounts can be held by more than one customer and a customer can have more than one account.
- A loan originates at a particular branch and can be held by one or more customers

Identify the relationship among the entities along with the mapping cardinalities, keys in the E.R. diagram. Construct appropriate tables for E-R diagram designed with above requirements.

OR

Q2) a) Explain the concept of candidate key and primary key, foreign key. Identify above listed key for the following schema: **[6]**

Person (driver_id, name, address, contactno)

Car(licence, model, year)

Owns (driver_id, licence)

b) Draw architecture of DBMS system and explain function of following components: **[9]**

i) Storage manager

ii) Query Processor

Q3) a) What is view and how to create it? Can you update view? If yes, how? If not, why not? **[5]**

b) Defined stored procedure. Explain the creating and calling stored procedure with example. **[5]**

c) Consider following schema. **[5]**

Student_fee_details (rollno, name, fee_deposited, date)

Write a trigger to preserve old values of student fee details before updating in the table.

OR

Q4) a) Consider the following schemes **[6]**

Supplier(SNO, Sname, Status, City)

Parts (PNO, Pname, Color, Weight, City)

Shipments(SNO,PNO,QTY)

Write SQL queries for the following:

i) Find shipment information (SNO, Sname, PNO, Pname, QTY) for those having quantity less than 157.

ii) List SNO, Sname, PNO, Pname for those suppliers who made shipments of parts whose quantity is larger than the average quantity

iii) Find aggregate quantity of PNO 1692 of color green for which shipments made by supplier number who residing Mumbai.

b) What is an index? What are the advantages and disadvantages of using index on a table? **[4]**

c) What is a trigger? How to create it? Discuss various types of triggers.**[5]**

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